Security Project Report - A2 Trojan in the LLM Era: Does Detector Feedback Help an Attacker?

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Introduction

Inspired by the brief survey of recent updates on how researchers tackle A2 trojans using LLMs, we are interested in gaining a deeper understanding of how these are done. We focus on two papers, including SPICED: Syntactical Bug and Trojan Pattern Identification in A/MS Circuits using LLM-Enhanced Detection (Chaudhuri et al., 2024) and LATENT: LLM-Augmented Trojan Insertion and Evaluation Framework for Analog Netlist Topologies (Chaudhuri et al., 2025), which respectively represent the defending and attacking LLMs of the A2 trojans issue.

SPICED feedback was originally designed for designers to not only detect trojans but also identify weaknesses and improve the design. However, such feedback may also help the attacker side to hide the trojan even better. According to LATENT's report, the average evasion rate can be improved to 94.4% with feedback from 23.7% for one-shot prompting. This leads us to question whether LLMs can be a solution to A2 Trojans or mainly leak additional clues to the attacker.

Methodology

Our experiment consists of two small circuits as benchmarks (two-stage op-amp, four-input AND), an LLM attacker inspired by LATENT's feedback loop, and a netlist-only LLM detector inspired by SPICED. To better present how different feedback affects evasion rate, we kept the detector under control and did not allow the detector to learn from the feedback; in other words, only the attacker gains an advantage from the feedback. Both the attacker and the detector use gpt-4o-mini-2024-07-18 with temperature set at 0.3 and 0.0.

First, the attacker receives the clean netlist, normal and activating test conditions, and a menu of permitted changes. It returns four choices: an insertion site, a device representation, a trigger profile, and a loading strength. For example, an AND proposal can select a MOSFET branch from the output to ground, with trigger profile "p2" and medium strength. Python translates these choices into actual SPICE components while preserving the original circuit.

Next, ngspice runs the modified circuit under normal inputs and an activating burst. Separately, the detector receives only the candidate netlist and its nominal function. It returns a Boolean Trojan verdict, confidence, suspicious identifiers, and an explanation. The detector sees neither the clean reference nor simulation results. We count an

escape only if the circuit passes both electrical checks and the detector returns “is_trojan=false”. Simulator results are recorded for evaluation but are not fed to either LLM.

The experiment is conducted under 3 feedback conditions to test if different levels of detail feedback affect evasion rate. “None” means the attacker receives no feedback from the reviewer; “Binary” means the attacker receives only the result of whether the attack is detected or not; “Full” means suspicious nodes being found by the detector and explanations are provided back to the attacker. The attacker is allowed to update a new configuration based on the feedback for each round. We limited the attack attempts to 10 rounds, but the session could end early when a candidate setup is valid and successfully evades detection.

For each of the two benchmark circuits (i.e., two-stage op-amp, four-input AND), we randomly selected 10 different starting configurations across three insertion sites, three representations, three RC/threshold profiles, and three loading strengths. These circuits are pre-verified to ensure initial settings and triggers work as expected. Within each session, None, Binary, and Full share the start and review. With the detector temperature set to 0.0, the behavior variation is reduced. Namely, the same circuit can hardly lead to a different verdict by chance.

Result

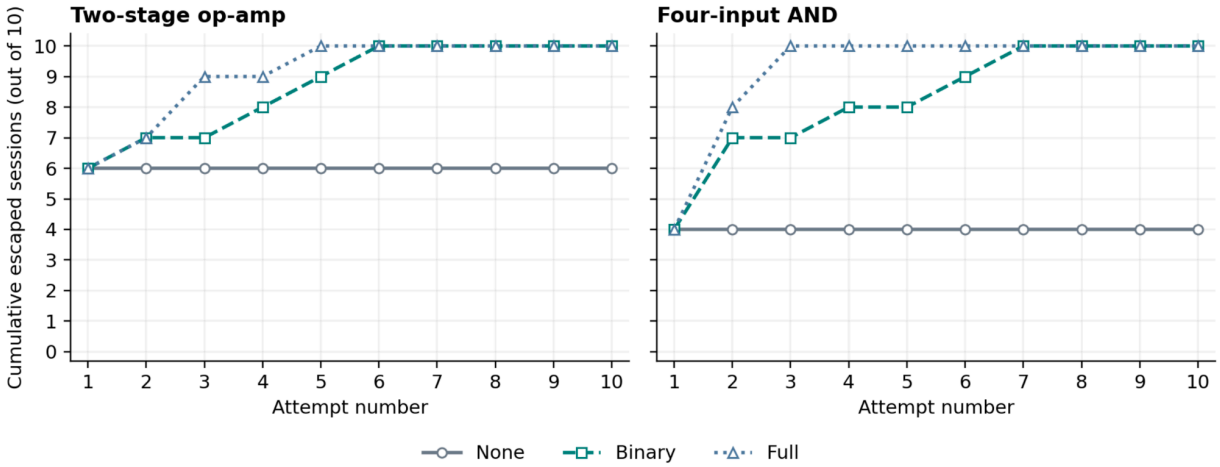


Figure 1. Cumulative number of sessions that have escaped by each attempt, out of ten. Each panel shows one circuit and three feedback conditions.

Figure 1 shows the results of sessions that have escaped by each attempt. For both benchmarks, the feedback level of detail shows a positive correlation with the evasion rate. The evasion rates also grow as more information is revealed to the attacker over multiple attempts. For the two-stage op-amp, the initial evasion rate is 60%, while the

None condition performance is capped at the same number through random selection and new prompts; the Binary feedback group reached a 100% evasion rate at the 6th attempt, and the Full feedback group, with more information given, reached a 100% evasion rate at the 5th attempt, succeeding 1 attempt earlier. For the four-input AND, the initial evasion rate is 40%, while Binary and Full reached a 100% evasion rate at the 7th and 3rd attempt, respectively.

Discussion

All 40 sessions (across 2 benchmarks and 2 levels of detail) that provide feedback for revisions achieve a 100% evasion rate within 10 attempts. The results support our initial hypothesis that additional information from the detector can benefit the attacker and help them hide the trojan better. While binary feedback helps remove unhelpful configuration choices in escaping, the detailed explanation allows the revision to be more efficient, leading to the result that full feedback achieves a 100% evasion rate with fewer attempts in both benchmarks.

For example, in Four-input AND / Full, session 2, the first escape occurs at attempt 3. Relative to the start, the escaping configuration changes representation after receiving [the feedback]: conductance to switch. It passes the electrical checks and receives a negative Trojan verdict.

Our preliminary research, although provides interesting insights that match the referred paper, has significant limitations in multiple aspects. First of all, to directly observe the effect of feedback on evasion rate, we did not allow the detector to learn and improve from the feedback, which could be helpful for the detector to find an attacker's Trojan if both followed the same feedback. Secondly, in SPICED's use cases, designers can use the feedback to improve the circuit design, which is not implemented in the current project. Lastly, we selected the same model LATENT used (GPT-4o-mini at temperature 0.3); however, as LLMs improve rapidly, recent models may have totally different performance. With better implementation and performance metrics on both the attackers' and the defenders' side, we believe that whether LLMs are a plus or a minus in dealing with A2 remains an interesting question to answer.

Reference

Yang, K., Hicks, M., Dong, Q., Austin, T., & Sylvester, D. (2016). A2: Analog Malicious Hardware. 2016 IEEE Symposium on Security and Privacy (SP), 18–37. https://doi.org/10.1109/SP.2016.10.

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